Prosit PL/SQL

1a)

```
SQL> connect hr
Enter password:
Connected.
SQL> set serveroutput on;
SQL> begin
  2  dbms_output.put_line('la date daujourdhui est:' ||sysdate);
  3  end;
  4  /
la date daujourdhui est:01-APR-26

PL/SQL procedure successfully completed.

SQL>
```

b)

```
SQL> declare
  2  a number :=44.5;
  3  b NUMBER := 61.8;
  4  BEGIN
  5      DBMS_OUTPUT.PUT_LINE('Somme = ' || (a + b));
  6  END;
  7  /
Somme = 106.3

PL/SQL procedure successfully completed.

SQL>
```

c)

```
SQL> begin
  2  for i in 1..10 loop
  3  dbms_output.put_line(i);
  4  end loop;
  5  dbms_output.put_line('********');
  6  for i in reverse 0..50 loop
  7  dbms_output.put_line(i*2);
  8  end loop;
  9  end;
 10  /
1
2
3
4
5
6
7
8
9
10
********
100
98
96
94
92
90
88
86
84
82
80
78
76
```

d)

```
SQL> DECLARE
  2      a NUMBER := &a;
  3      b NUMBER := &b;
  4  BEGIN
  5      DBMS_OUTPUT.PUT_LINE('somme = ' || (a + b));
  6      DBMS_OUTPUT.PUT_LINE('produit = ' || (a * b));
  7  END;
  8  /
Enter value for a: 3
old   2:     a NUMBER := &a;
new   2:     a NUMBER := 3;
Enter value for b: 5
old   3:     b NUMBER := &b;
new   3:     b NUMBER := 5;
somme = 8
produit = 15

PL/SQL procedure successfully completed.

SQL>
```

e)

```
SQL> DECLARE
  2      A NUMBER := &A;
  3      B NUMBER := &B;
  4      C NUMBER := &C;
  5  BEGIN
  6      IF A > B AND B > C THEN
  7          DBMS_OUTPUT.PUT_LINE('A > B > C');
  8      ELSIF A > C AND C > B THEN
  9          DBMS_OUTPUT.PUT_LINE('A > C > B');
 10      ELSIF B > A AND A > C THEN
 11          DBMS_OUTPUT.PUT_LINE('B > A > C');
 12      ELSE
 13          DBMS_OUTPUT.PUT_LINE('Autre ordre');
 14      END IF;
 15  END;
 16  /
Enter value for a: 20
old   2:     A NUMBER := &A;
new   2:     A NUMBER := 20;
Enter value for b: 40
old   3:     B NUMBER := &B;
new   3:     B NUMBER := 40;
Enter value for c: 60
old   4:     C NUMBER := &C;
new   4:     C NUMBER := 60;
Autre ordre

PL/SQL procedure successfully completed.

SQL>
```

f)

```
SQL> DECLARE
  2      A NUMBER := &A;
  3      fact NUMBER := 1;
  4  BEGIN
  5      FOR i IN 1..A LOOP
  6          fact := fact * i;
  7      END LOOP;
  8
  9      DBMS_OUTPUT.PUT_LINE('Factoriel = ' || fact);
 10  END;
 11  /
Enter value for a: 20
old   2:     A NUMBER := &A;
new   2:     A NUMBER := 20;
Factoriel = 2432902008176640000

PL/SQL procedure successfully completed.

SQL>
```

2a)

```
SQL> DECLARE
         v_nom employees.last_name%TYPE;
     BEGIN
         SELECT last_name INTO v_nom
         FROM employees
         WHERE employee_id = 110;

         DBMS_OUTPUT.PUT_LINE('Nom : ' || v_nom);
     END;
     /
Nom : Chen

PL/SQL procedure successfully completed.

SQL>
```

b)

```
SQL> DECLARE
         nb NUMBER;
     BEGIN
         SELECT COUNT(*) INTO nb
         FROM employees
         WHERE department_id = 50;

         DBMS_OUTPUT.PUT_LINE('Nombre employés : ' || nb);
     END;
     /
Nombre employés : 45

PL/SQL procedure successfully completed.

SQL>
```

c)

```
SQL> DECLARE
         sal_min NUMBER;
         sal_max NUMBER;
     BEGIN
         SELECT MIN(salary), MAX(salary)
         INTO sal_min, sal_max
         FROM employees
         WHERE department_id = 50;

         DBMS_OUTPUT.PUT_LINE('Min = ' || sal_min);
         DBMS_OUTPUT.PUT_LINE('Max = ' || sal_max);
     END;
     /
Min = 2100
Max = 8200

PL/SQL procedure successfully completed.

SQL>
```

3a)

```
SQL> declare
s_ancien employees.salary%type;
s_recent employees.salary%type;
ecart number;

begin

select salary into s_ancien from employees where department_id=50 and hire_date=(select min(hire_date) from employees where department_id=50);


select salary into s_recent from employees where department_id=50 and hire_date=(select max(hire_date) from employees where department_id=50);
 ecart:=s_ancien-s_recent;

DBMS_OUTPUT.PUT_LINE('Salaire employé le plus ancien : ' || s_ancien);
DBMS_OUTPUT.PUT_LINE('Salaire employé le plus récent : ' || s_recent);
DBMS_OUTPUT.PUT_LINE('Ecart : ' || ecart);

end;
/
Salaire employé le plus ancien : 7900
Salaire employé le plus récent : 2200
Ecart : 5700

PL/SQL procedure successfully completed.

SQL>
```

b)

```
SQL> declare
        d1 number;
        d2 number;
        begin
         select avg(salary) into d1 from employees where department_id=50;
         select avg(salary) into d2 from employees where department_id=30;
         IF d1 > d2 THEN    DBMS_OUTPUT.PUT_LINE('Moyenne dep 50 > dep 30');  ELSIF d1 < d2 THEN    DBMS_OUTPUT.PUT_LINE('Moyenne dep 50 < dep 30');  ELSE    DBMS_OUTPUT.PUT_LINE('Les deux moyennes sont égales');  END IF;
           DBMS_OUTPUT.PUT_LINE('d1 = ' || d1);  DBMS_OUTPUT.PUT_LINE('d2 = ' || d2);
         end;
   /
Moyenne dep 50 < dep 30
d1 = 3475.55555555555555555555555555555555556
d2 = 4150

PL/SQL procedure successfully completed.

SQL>
```